TASK 2: Network Security & Scanning

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1. Objective

The objective of this task is to practise reconnaissance, host discovery, port and service scanning, packet analysis, vulnerability assessment, and basic firewall concepts in an authorised virtual lab. Testing is limited to the isolated Metasploitable2 target at 192.168.56.101.

2. Lab Environment and Scope

- Attacker/analysis system: Kali Linux running in Oracle VirtualBox.
- Target system: Metasploitable2 virtual machine.
- Target IP observed in Nmap output: 192.168.56.101.
- Tools used or checked: Nmap, Wireshark, PostgreSQL/Greenbone components, and iptables.
- Scope: the user's isolated lab systems only. No public or third-party systems were included in the scan.

The full Nmap output shows a scan using Nmap 7.99 with TCP SYN scanning, service/version detection, and OS detection enabled. The scan completed in approximately 206.80 seconds.

3. Reconnaissance and Host Discovery

3.1 Passive reconnaissance

WHOIS and nslookup were run against example.com for practice. The commands were attempted and output files were intended to be saved in the reports directory. Attach the actual command outputs/screenshots as evidence before final submission; this report does not assume specific WHOIS registration details or DNS answers.

3.2 Active reconnaissance

A ping sweep was attempted against the isolated lab subnet 192.168.56.0/24. Review the saved ping_sweep.txt output and insert a screenshot confirming the discovered hosts.

4. Nmap Port and Service Scanning

The Nmap full scan identified the target as up and listed 23 open TCP ports. It reported 977 filtered TCP ports that did not respond. The scan did not conclusively identify the operating system; Nmap warned that OS fingerprinting was unreliable because it could not find at least one open and one closed port.

4.1 Selected observed services

Port	Service	Nmap-reported version/banner
21/tcp	FTP	vsftpd 2.3.4
22/tcp	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23/tcp	Telnet	Linux telnetd

25/tcp	SMTP	Postfix smtpd
53/tcp	DNS	ISC BIND 9.4.2
80/tcp	HTTP	Apache httpd 2.2.8 (Ubuntu), DAV/2
111/tcp	RPC	rpcbind 2
139/tcp	SMB	Samba smbd 3.X–4.X
445/tcp	SMB	Samba smbd 3.X–4.X
512/tcp	exec	netkit-rsh rexecd
513/tcp	login	Service version uncertain
514/tcp	shell	Netkit rshd
1099/tcp	Java RMI	GNU Classpath grmiregistry
1524/tcp	shell	Metasploitable root shell (Nmap service label)
2049/tcp	NFS	NFS service
2121/tcp	FTP	ProFTPD 1.3.1
3306/tcp	MySQL	MySQL 5.0.51a-3ubuntu5
5432/tcp	PostgreSQL	PostgreSQL 8.3.0–8.3.7
5900/tcp	VNC	VNC protocol 3.3
6000/tcp	X11	Access denied
6667/tcp	IRC	UnrealIRCd
8009/tcp	AJP	AJP13
8180/tcp	HTTP	Apache Tomcat/Coyote JSP engine 1.1

5. Preliminary Security Analysis

- Unencrypted remote access: Telnet is not encrypted and can expose authentication information on the network. Prefer SSH for secure remote administration.
- Legacy software: the detected banners identify several old service versions. Verify exact package versions and applicable advisories before concluding that a particular vulnerability is present.
- Multiple exposed services: numerous network services increase the attack surface. Disable unnecessary services and restrict access with firewall rules.
- Database exposure: MySQL and PostgreSQL were reachable on ports 3306 and 5432. Database access should be limited to authorised hosts and protected with strong authentication.
- FTP: the detected vsftpd 2.3.4 banner warrants validation against reliable vulnerability references. A version banner alone does not prove exploitability.
- Remote administration and file sharing: SMB, VNC, RPC, NFS, and Java RMI should be reviewed for necessity, authentication, patch level, and network exposure.

These are preliminary observations based on service discovery, not confirmed vulnerability findings. A dedicated vulnerability scan and safe service-specific verification are required before assigning severity or claiming exploitability.

6. Wireshark Packet Analysis

The task instructions require capture and analysis of HTTP, FTP, and DNS traffic. DNS capture activity was attempted, and HTTP/FTP capture steps were discussed. Before submission, insert evidence from the actual saved capture files and confirm which protocols and packets were successfully observed. Do not claim that credentials were recovered or that a particular protocol was captured unless the packet evidence supports it.

- DNS: document the queried name, query type, response, and resolved address if visible.
- HTTP: document the request method, host/URI, and response status where visible.
- FTP: document whether commands and server responses are visible; explain the security risks of plaintext FTP.
- Evidence files/screenshots: insert filenames and screenshots from the actual captures.

7. Basic Firewall Verification

The iptables INPUT and OUTPUT chains were inspected. The captured INPUT/OUTPUT listing initially showed ACCEPT policies with no rules displayed. Later firewall-rule activity was discussed, but the final effective rule and its counter must be verified from the saved firewall_rules.txt output before stating that a port is blocked.

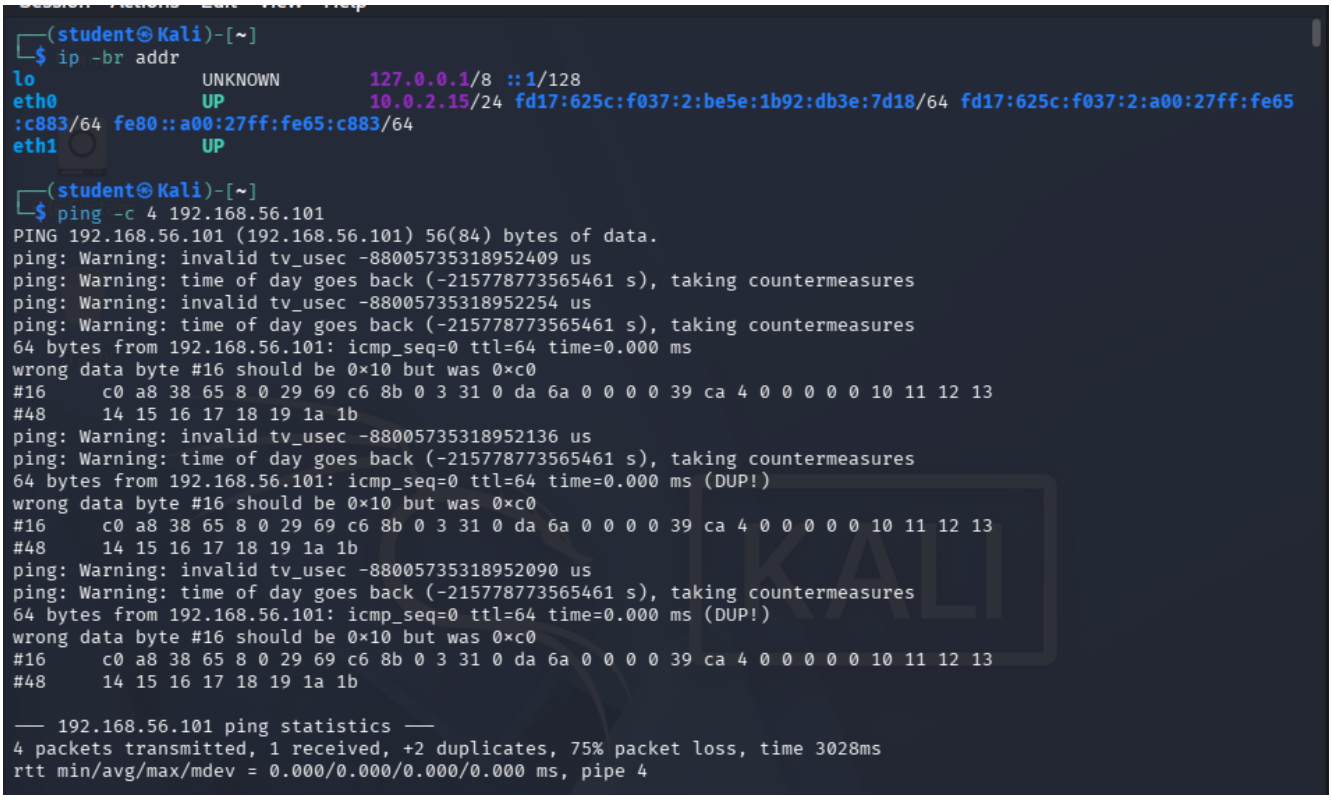
Evidence to attach: output of `sudo iptables -L INPUT -n -v --line-numbers` and the final saved `firewall_rules.txt`. If a DROP rule is present, document its exact chain, protocol, destination port, and action.

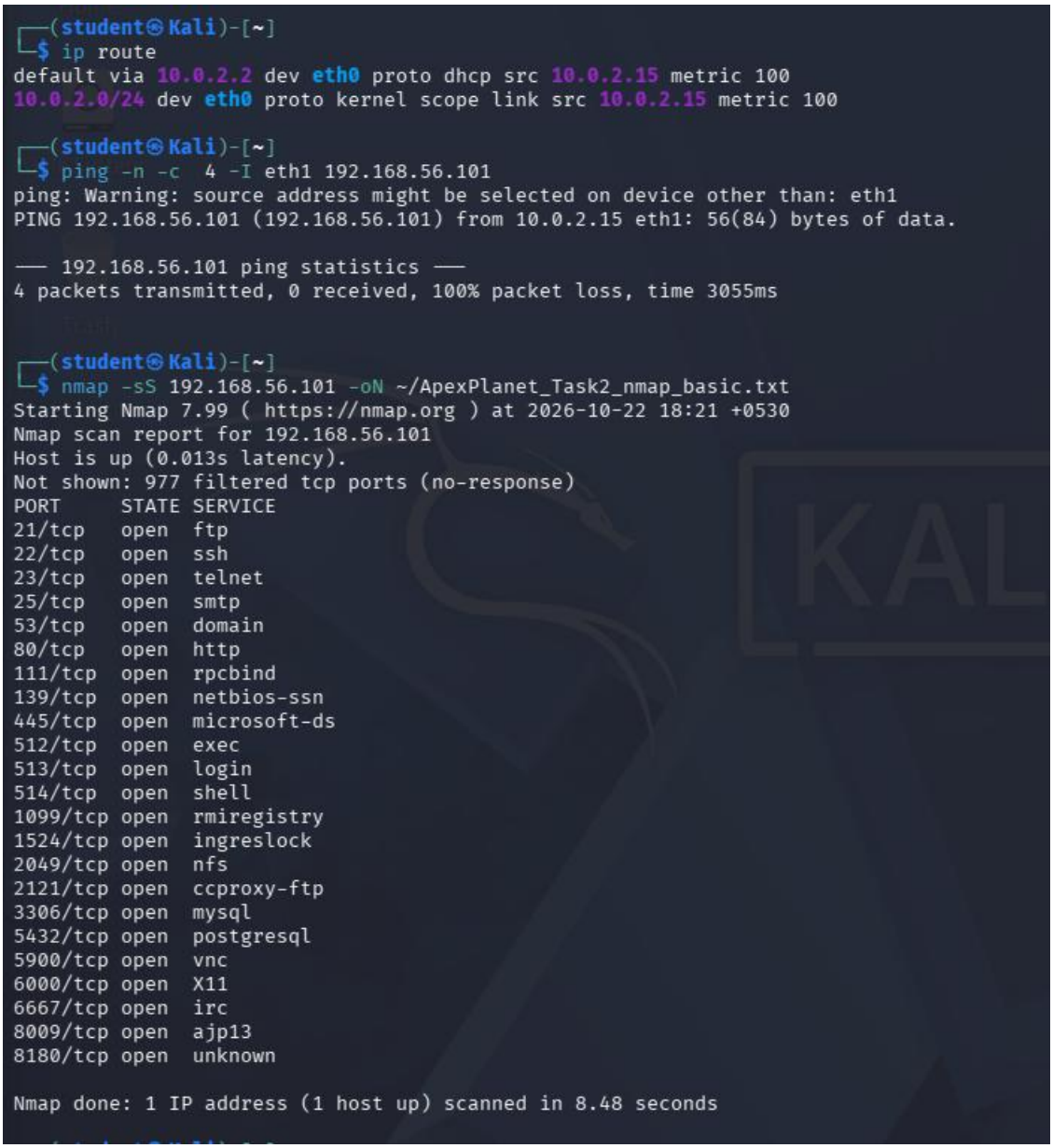
8. Vulnerability Scanning Status

Greenbone/OpenVAS components were installed, including gvmd, ospd-openvas, greenbone-feed-sync, and the PostgreSQL pg-gvm extension. The ospd-openvas service was observed running, but the gvmd manager repeatedly timed out. The PostgreSQL database listing contained only postgres, template0, and template1; a dedicated gvmd database was not present. Therefore, a successful Greenbone vulnerability scan and exported vulnerability report have not yet been confirmed. This deliverable remains PENDING and must not be represented as completed.

Next action: repair Greenbone using the supported setup process for the installed Kali packages, or use an approved alternative scanner in the same authorised lab. Export the actual findings and add them to this report when available.

9. Practical Activities Screenshots




┌──(student㉿Kali)-[~]
└─$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
┌──(student㉿Kali)-[~]
└─$ ping -n -c 4 -I eth1 192.168.56.101
ping: Warning: source address might be selected on device other than: eth1
PING 192.168.56.101 (192.168.56.101) from 10.0.2.15 eth1: 56(84) bytes of data.
— 192.168.56.101 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3055ms
┌──(student㉿Kali)-[~]
└─$ nmap -sS 192.168.56.101 -oN ~/ApexPlanet_Task2_nmap_basic.txt
Starting Nmap 7.99 (https://nmap.org) at 2026-10-22 18:21 +0530
Nmap scan report for 192.168.56.101
Host is up (0.013s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
Nmap done: 1 IP address (1 host up) scanned in 8.48 seconds

```
┌──(student㉿Kali)-[~]
└─$ cat ~/ApexPlanet_Task2_nmap_basic.txt
# Nmap 7.99 scan initiated Thu Oct 22 18:21:56 2026 as: /usr/lib/nmap/nmap --privileged -sS -oN /home/student/ApexP
lanet_Task2_nmap_basic.txt 192.168.56.101
Nmap scan report for 192.168.56.101
Host is up (0.013s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown

# Nmap done at Thu Oct 22 18:22:04 2026 -- 1 IP address (1 host up) scanned in 8.48 seconds
```

```
┌──(student㉿Kali)-[~]
└─$ nmap -sV 192.168.56.101 -oN ~/ApexPlanet_Task2_nmap_versions.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-22 18:26 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13?
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux
_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 193.34 seconds
```

```
└─$ cat ~/ApexPlanet_Task2_nmap_versions.txt
# Nmap 7.99 scan initiated Thu Oct 22 18:26:09 2026 as: /usr/lib/nmap/nmap --privileged -sV -oN /home/student/ApexP
lanet_Task2_nmap_versions.txt 192.168.56.101
Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13?
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux
_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Thu Oct 22 18:29:23 2026 -- 1 IP address (1 host up) scanned in 193.34 seconds

┌──(student㉿Kali)-[~]
└─$ nmap -sS -sV -O 192.168.56.101 -oN ~/ApexPlanet_Task2_nmap_full.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-22 18:32 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
```

```
┌──(student㉿Kali)-[~]
└─$ cat ~/ApexPlanet_Task2_nmap_full.txt

# Nmap 7.99 scan initiated Thu Oct 22 18:32:17 2026 as: /usr/lib/nmap/nmap --privileged -sS -sV -O -oN /home/studen
t/ApexPlanet_Task2_nmap_full.txt 192.168.56.101
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13?
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode ne
twork gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux
_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Thu Oct 22 18:35:44 2026 -- 1 IP address (1 host up) scanned in 206.80 seconds
```

```
┌──(student㉿Kali)-[~]
└─$ cat ~/ApexPlanet_Task2_nmap_udp.txt
# Nmap 7.99 scan initiated Thu Oct 22 18:40:05 2026 as: /usr/lib/nmap/nmap -sU --top-ports 100 -oN /home/student/Ap
exPlanet_Task2_nmap_udp.txt 192.168.56.101
Nmap scan report for 192.168.56.101
Host is up (0.0054s latency).
Not shown: 96 open|filtered udp ports (no-response)
PORT     STATE SERVICE
53/udp   open  domain
111/udp  open  rpcbind
137/udp  open  netbios-ns
2049/udp open  nfs

# Nmap done at Thu Oct 22 18:40:08 2026 -- 1 IP address (1 host up) scanned in 2.28 seconds

┌──(student㉿Kali)-[~]
└─$ mkdir -p ~/ApexPlanet_Task2/reprts

┌──(student㉿Kali)-[~]
└─$ cp ~/ApexPlanet_Task2_nmap_basic.txt ~/ApexPlanet_Task2/reports/
cp: cannot create regular file '/home/student/ApexPlanet_Task2/reports/': Not a directory

┌──(student㉿Kali)-[~]
└─$ ls -ld ~/ApexPlanet_Task2 ~/ApexPlanet_Task2/reports
ls: cannot access '/home/student/ApexPlanet_Task2/reports': No such file or directory
drwxrwxr-x 3 student student 4096 Oct 22 18:42 /home/student/ApexPlanet_Task2

┌──(student㉿Kali)-[~]
└─$ mkdir -p ~/ApexPlanet_Task2/reports

┌──(student㉿Kali)-[~]
└─$ ls -ld ~/ApexPlanet_Task2/reports
drwxrwxr-x 2 student student 4096 Oct 22 18:47 /home/student/ApexPlanet_Task2/reports

┌──(student㉿Kali)-[~]
└─$ cp ~/ApexPlanet_Task2_nmap_basic.txt ~/ApexPlanet_Task2/reports/

┌──(student㉿Kali)-[~]
└─$ cp ~/ApexPlanet_Task2_nmap_versions.txt ~/ApexPlanet_Task2/reports/

┌──(student㉿Kali)-[~]
└─$ cp ~/ApexPlanet_Task2_nmap_full.txt ~/ApexPlanet_Task2/reports/


┌──(student㉿Kali)-[~]
└─$ cp ~/ApexPlanet_Task2_nmap_udp.txt ~/ApexPlanet_Task2/reports/
```

```
┌──(student㉿Kali)-[~]
└─$ ls -lh ~/ApexPlanet_Task2/reports/
total 16K
-rw-rw-r-- 1 student student  895 Oct 22 18:49 ApexPlanet_Task2_nmap_basic.txt
-rw-rw-r-- 1 student student 2.2K Oct 22 18:51 ApexPlanet_Task2_nmap_full.txt
-rw-r--r-- 1 student student  482 Oct 22 18:51 ApexPlanet_Task2_nmap_udp.txt
-rw-rw-r-- 1 student student 1.7K Oct 22 18:50 ApexPlanet_Task2_nmap_versions.txt

┌──(student㉿Kali)-[~]
└─$ command -v gvm

┌──(student㉿Kali)-[~]
└─$ command -v openvas
/usr/sbin/openvas

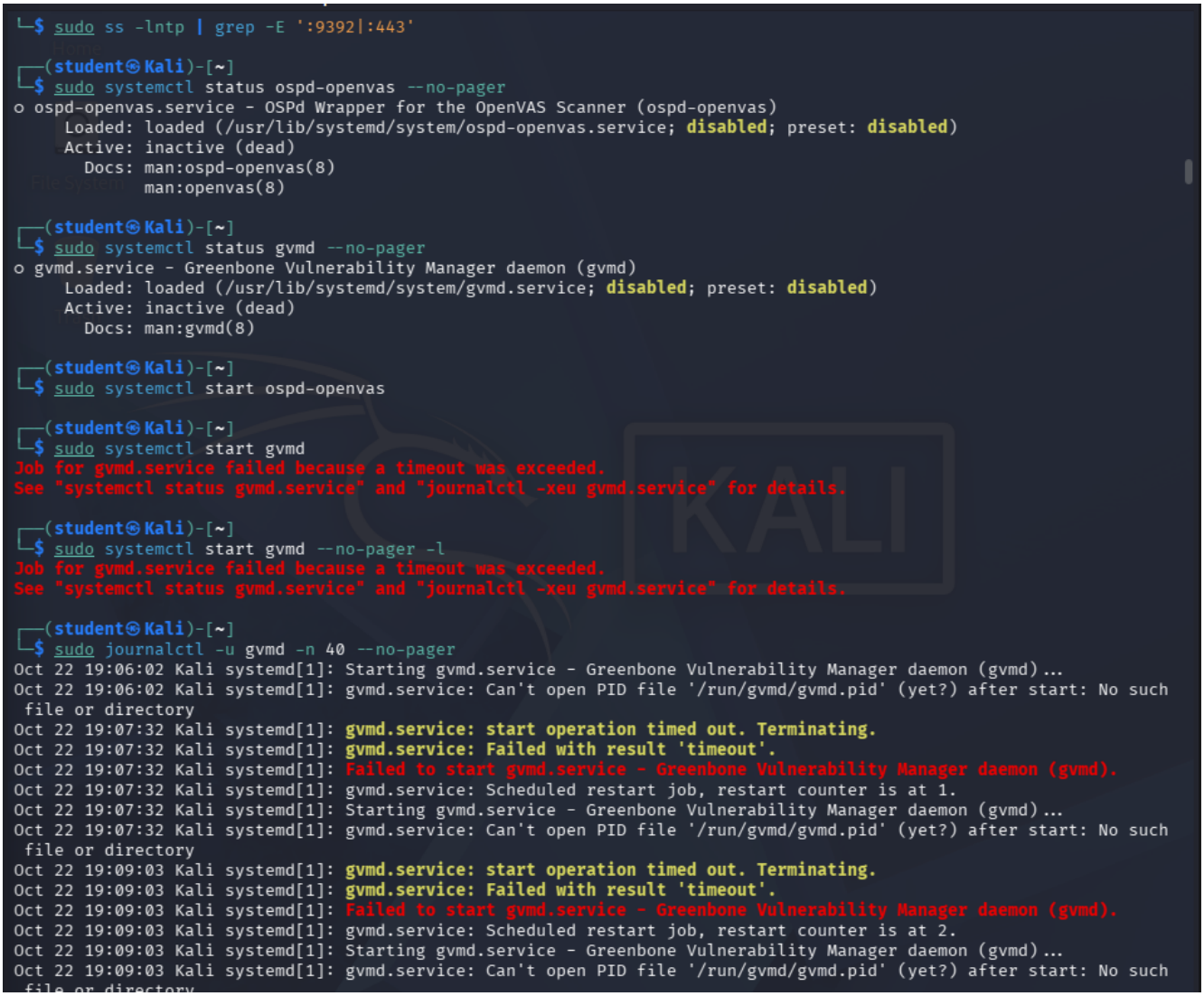
┌──(student㉿Kali)-[~]
└─$ sudo gvm-check-setup
sudo: gvm-check-setup: command not found

┌──(student㉿Kali)-[~]
└─$ ls -lh /usr/sbin/openvas
-rwxr-xr-x 1 root root 83K Apr 28 19:26 /usr/sbin/openvas

┌──(student㉿Kali)-[~]
└─$ command -v gvmd
/usr/sbin/gvmd

┌──(student㉿Kali)-[~]
└─$ command -v gsad

┌──(student㉿Kali)-[~]
└─$ dpkg -l | grep -Ei 'openvas|gvm|greenbone'
ii  greenbone-feed-sync                  25.2.0-2                                all          New script for syn
cing the Greenbone Community Feed (Python 3)
ii  gvmd                                 26.24.0-1                               amd64        Manager Module of
Greenbone Vulnerability Manager
ii  gvmd-common                          26.24.0-1                               all          architecture indep
endent files for Greenbone Vulnerability Manager
ii  libgvm22t64:amd64                    22.41.0-1                               amd64        remote network sec
urity auditor - shared libraries
ii  openvas-scanner                      23.45.1-1                               amd64        remote network sec
urity auditor - scanner
ii  ospd-openvas                         22.10.0-1                               all          OSP server impleme
ntation to allow GVM to remotely control an OpenVAS Scanner
ii  postgresql-18-pg-gvm                 22.6.15-1                               amd64        PostgreSQL extensi
on for ical object manipulation
ii  python3-gvm                          26.11.0-1                               all          Greenbone Vulnerab
ility Management Python Library (Python 3)
```


└─$ sudo ss -lntp | grep -E ':9392|:443'
┌──(student㉿Kali)-[~]
└─$ sudo systemctl status ospd-openvas --no-pager
o ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd-openvas)
Loaded: loaded (/usr/lib/systemd/system/ospd-openvas.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:ospd-openvas(8)
man:openvas(8)
┌──(student㉿Kali)-[~]
└─$ sudo systemctl status gvmd --no-pager
o gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)
Loaded: loaded (/usr/lib/systemd/system/gvmd.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:gvmd(8)
┌──(student㉿Kali)-[~]
└─$ sudo systemctl start ospd-openvas
┌──(student㉿Kali)-[~]
└─$ sudo systemctl start gvmd
Job for gvmd.service failed because a timeout was exceeded.
See "systemctl status gvmd.service" and "journalctl -xeu gvmd.service" for details.
┌──(student㉿Kali)-[~]
└─$ sudo systemctl start gvmd --no-pager -l
Job for gvmd.service failed because a timeout was exceeded.
See "systemctl status gvmd.service" and "journalctl -xeu gvmd.service" for details.
┌──(student㉿Kali)-[~]
└─$ sudo journalctl -u gvmd -n 40 --no-pager
Oct 22 19:06:02 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 22 19:06:02 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such file or directory
Oct 22 19:07:32 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 22 19:07:32 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 22 19:07:32 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 22 19:07:32 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 1.
Oct 22 19:07:32 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 22 19:07:32 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such file or directory
Oct 22 19:09:03 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 22 19:09:03 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 22 19:09:03 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 22 19:09:03 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 2.
Oct 22 19:09:03 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 22 19:09:03 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such file or directory

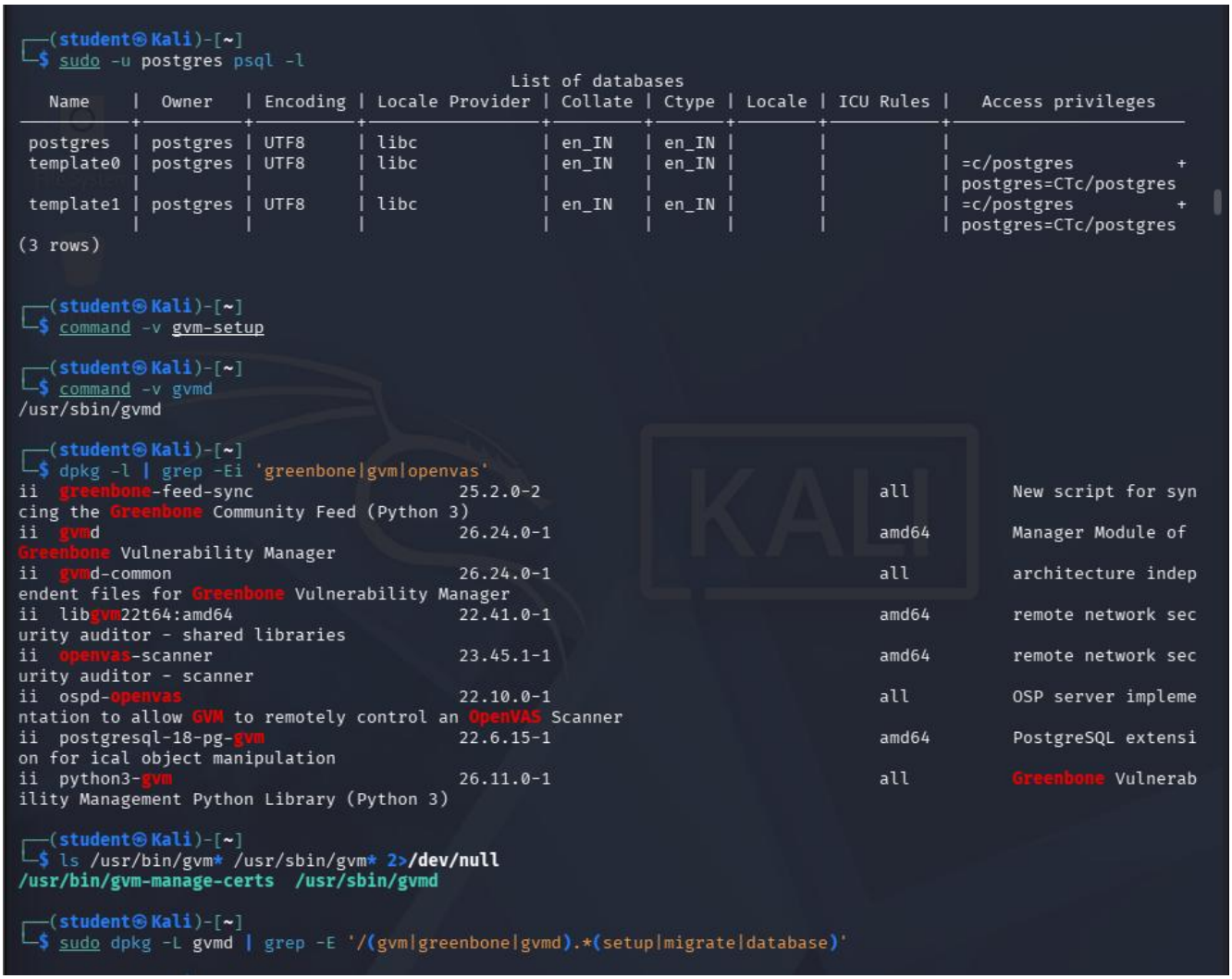
```
┌──(student㉿Kali)-[~]
└─$ sudo systemctl status postgresql --no-pager
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/usr/lib/systemd/system/postgresql.service; disabled; preset: disabled)
     Active: active (exited) since Thu 2026-10-22 19:06:02 IST; 5min ago
 Invocation: 4d26d01ea13949b982229f343408b089
    Process: 33507 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 33507 (code=exited, status=0/SUCCESS)
   Mem peak: 2M
        CPU: 6ms

Oct 22 19:06:02 Kali systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Oct 22 19:06:02 Kali systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.

┌──(student㉿Kali)-[~]
└─$ sudo systemctl status ospd-openvas --no-pager
● ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd-openvas)
     Loaded: loaded (/usr/lib/systemd/system/ospd-openvas.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-10-22 19:05:34 IST; 6min ago
 Invocation: 768fb1cc437e4e258fda5c047a0a13c0
       Docs: man:ospd-openvas(8)
             man:openvas(8)
    Process: 33176 ExecStart=/usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/gvm/ospd-l
ogging.conf (code=exited, status=0/SUCCESS)
   Main PID: 33194 (ospd-openvas)
      Tasks: 5 (limit: 4618)
     Memory: 68.6M (peak: 110.9M)
        CPU: 2.820s
     CGroup: /system.slice/ospd-openvas.service
             ├─33194 /usr/bin/python3 /usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/…
             └─33196 /usr/bin/python3 /usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/…

Oct 22 19:05:33 Kali systemd[1]: Starting ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (osp…nvas)...
Oct 22 19:05:34 Kali systemd[1]: Started ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd…penvas).
Hint: Some lines were ellipsized, use -l to show in full.

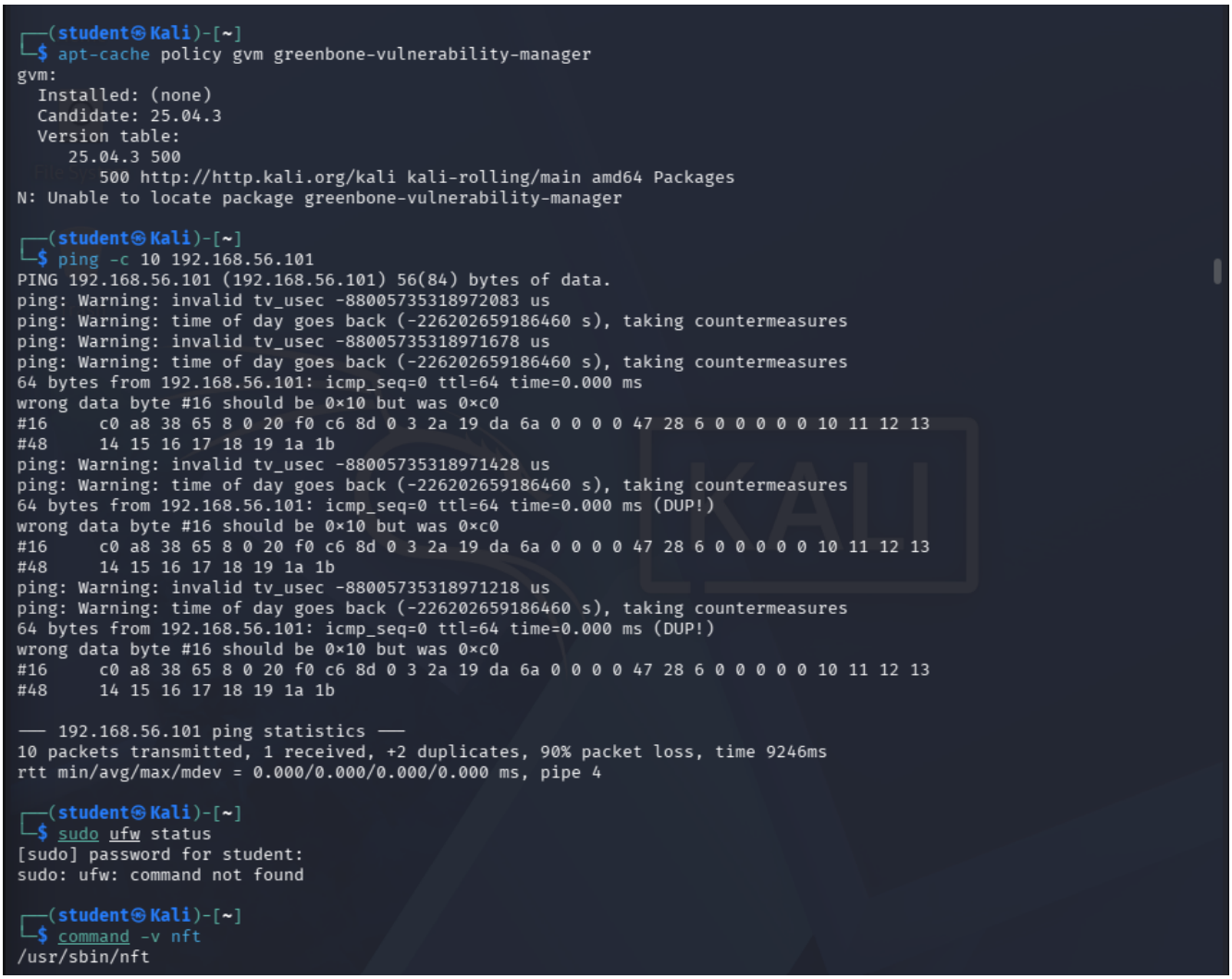
┌──(student㉿Kali)-[~]
└─$ sudo pg_lsclusters
Ver Cluster Port Status Owner    Data directory              Log file
18  main    5432 online postgres /var/lib/postgresql/18/main /var/log/postgresql/postgresql-18-main.log
```


┌──(student㉿Kali)-[~]
└─$ sudo -u postgres psql -l
List of databases
Name | Owner | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules | Access privileges
postgres | postgres | UTF8 | libc | en_IN | en_IN | | |
template0 | postgres | UTF8 | libc | en_IN | en_IN | | | =c/postgres + postgres=CTc/postgres
template1 | postgres | UTF8 | libc | en_IN | en_IN | | | =c/postgres + postgres=CTc/postgres
(3 rows)
┌──(student㉿Kali)-[~]
└─$ command -v gvm-setup
┌──(student㉿Kali)-[~]
└─$ command -v gvmd
/usr/sbin/gvmd
┌──(student㉿Kali)-[~]
└─$ dpkg -l | grep -Ei 'greenbone|gvm|openvas'
ii greenbone-feed-sync 25.2.0-2 all New script for syncing the Greenbone Community Feed (Python 3)
ii gvmd 26.24.0-1 amd64 Manager Module of Greenbone Vulnerability Manager
ii gvmd-common 26.24.0-1 all architecture independent files for Greenbone Vulnerability Manager
ii libgvm22t64:amd64 22.41.0-1 amd64 remote network security auditor - shared libraries
ii openvas-scanner 23.45.1-1 amd64 remote network security auditor - scanner
ii ospd-openvas 22.10.0-1 all OSP server implementation to allow GVM to remotely control an OpenVAS Scanner
ii postgresql-18-pg-gvm 22.6.15-1 amd64 PostgreSQL extension for ical object manipulation
ii python3-gvm 26.11.0-1 all Greenbone Vulnerability Management Python Library (Python 3)
┌──(student㉿Kali)-[~]
└─$ ls /usr/bin/gvm* /usr/sbin/gvm* 2>/dev/null
/usr/bin/gvm-manage-certs /usr/sbin/gvmd
┌──(student㉿Kali)-[~]
└─$ sudo dpkg -L gvmd | grep -E '/(gvm|greenbone|gvmd).*(setup|migrate|database)'

```
└─$ gvmd --help
Usage:
  gvmd [OPTION…] - Manager of the Open Vulnerability Assessment System

Help Options:
  -h, --help                                  Show help options

Application Options:
  --affected-products-query-size=<number>     Sets the number of CVEs to process per query when updating the af
fected products. Defaults to 20000.
  --auth-timeout=<timeout>                    Sets the authentication timeout time for the cached authenticatio
n. Defaults to 15 minutes.
  --broker-address=<address>                  Sets the address for the publish-subscribe message (MQTT) broker.
 Defaults to localhost:1883. Set to empty to disable.
  --check-alerts                              Check SecInfo alerts.
  --client-watch-interval=<number>            Check if client connection was closed every <number> seconds. 0 t
o disable. Defaults to 1 seconds.
  --create-encryption-key                     Create a new credential encryption key, set it as the new default
 and exit. With no other options given, a 4096 bit RSA key is created.
  --create-scanner=<scanner>                  Create global scanner <scanner> and exit.
  --create-user=<username>                    Create admin user <username> and exit.
  -d, --database=<name>                       Use <name> as database for PostgreSQL.
  --db-host=<host>                            Use <host> as database host or socket directory for PostgreSQL.
  --db-port=<port>                            Use <port> as database port or socket extension for PostgreSQL.
  --db-user=<user>                            Use <user> as database user.
  --db-semaphore-timeout=<semaphore_timeout>  Use <semaphore_timeout> as sempahore timeout for PostgreSQL conne
ctions.
  --delete-scanner=<scanner-uuid>             Delete scanner <scanner-uuid> and exit.
  --delete-user=<username>                    Delete user <username> and exit.
  --dh-params=<file>                          Diffie-Hellman parameters file
  --disable-cmds=<commands>                   Disable comma-separated <commands>.
  --disable-encrypted-credentials             Do not encrypt or decrypt credentials.
  --disable-password-policy                   Do not restrict passwords to the policy.
  --disable-scheduling                        Disable task scheduling.
  --dump-vt-verification                      Dump the string the VTs verification hash is calculated from.
  --dump-asset-snapshot-counts                Dump the string the Asset snapshot counts are calculated from.
  --encryption-key-length=<length>            Set key length to <length> bits when creating a new RSA credentia
l encryption key. Defaults to DEFAULT_ENCRYPTION_KEY_LENGTH.
  --encryption-key-type=<type>                Use the key type <type> when creating a new credential encryption
 key. Currently only RSA is supported.
  --encrypt-all-credentials                   (Re-)Encrypt all credentials.
  --feed-lock-path=<path>                     Sets the path to the feed lock file.
  --feed-lock-timeout=<timeout>               Sets the number of seconds to retry for if the feed is locked in
contexts (like migration or rebuilds) that do not retry on their own (like automatic syncs). Defaults to 0 (no retr
y).
  -f, --foreground                            Run in foreground.
  --get-roles                                 List roles and exit.
  --get-scanners                              List scanners and exit.
  --get-users                                 List users and exit.
  --gnutls-priorities=<priorities-string>     Sets the GnuTLS priorities for the Manager socket.
```


┌──(student㉿Kali)-[~]
└─$ apt-cache policy gvm greenbone-vulnerability-manager
gvm:
Installed: (none)
Candidate: 25.04.3
Version table:
25.04.3 500
500 http://http.kali.org/kali kali-rolling/main amd64 Packages
N: Unable to locate package greenbone-vulnerability-manager
┌──(student㉿Kali)-[~]
└─$ ping -c 10 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
ping: Warning: invalid tv_usec -88005735318972083 us
ping: Warning: time of day goes back (-226202659186460 s), taking countermeasures
ping: Warning: invalid tv_usec -88005735318971678 us
ping: Warning: time of day goes back (-226202659186460 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.000 ms
wrong data byte #16 should be 0×10 but was 0×c0
#16 c0 a8 38 65 8 0 20 f0 c6 8d 0 3 2a 19 da 6a 0 0 0 0 47 28 6 0 0 0 0 0 10 11 12 13
#48 14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735318971428 us
ping: Warning: time of day goes back (-226202659186460 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0×10 but was 0×c0
#16 c0 a8 38 65 8 0 20 f0 c6 8d 0 3 2a 19 da 6a 0 0 0 0 47 28 6 0 0 0 0 0 10 11 12 13
#48 14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735318971218 us
ping: Warning: time of day goes back (-226202659186460 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0×10 but was 0×c0
#16 c0 a8 38 65 8 0 20 f0 c6 8d 0 3 2a 19 da 6a 0 0 0 0 47 28 6 0 0 0 0 0 10 11 12 13
#48 14 15 16 17 18 19 1a 1b
── 192.168.56.101 ping statistics ──
10 packets transmitted, 1 received, +2 duplicates, 90% packet loss, time 9246ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4
┌──(student㉿Kali)-[~]
└─$ sudo ufw status
[sudo] password for student:
sudo: ufw: command not found
┌──(student㉿Kali)-[~]
└─$ command -v nft
/usr/sbin/nft

```
┌──(student㉿Kali)-[~]
└─$ command -v firewall-cmd

┌──(student㉿Kali)-[~]
└─$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num   pkts bytes target     prot opt in     out     source               destination

┌──(student㉿Kali)-[~]
└─$ sudo iptables -L OUTPUT -n -v --line-numbers
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num   pkts bytes target     prot opt in     out     source               destination

┌──(student㉿Kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 80 -j DROP

┌──(student㉿Kali)-[~]
└─$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num   pkts bytes target     prot opt in     out     source               destination
1        0     0 DROP       tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:80

┌──(student㉿Kali)-[~]
└─$ ls -lh ~/ApexPlanet_Task2/
total 8.0K
drwxrwxr-x 2 student student 4.0K Oct 22 18:51 reports
drwxrwxr-x 2 student student 4.0K Oct 22 18:42 reprts

┌──(student㉿Kali)-[~]
└─$ ls -lh ~/ApexPlanet_Task2/reports
total 16K
-rw-rw-r-- 1 student student  895 Oct 22 18:49 ApexPlanet_Task2_nmap_basic.txt
-rw-rw-r-- 1 student student 2.2K Oct 22 18:51 ApexPlanet_Task2_nmap_full.txt
-rw-r--r-- 1 student student  482 Oct 22 18:51 ApexPlanet_Task2_nmap_udp.txt
-rw-rw-r-- 1 student student 1.7K Oct 22 18:50 ApexPlanet_Task2_nmap_versions.txt

┌──(student㉿Kali)-[~]
└─$ command -v whois
/usr/bin/whois

┌──(student㉿Kali)-[~]
└─$ command -v nslookup
/usr/bin/nslookup

┌──(student㉿Kali)-[~]
└─$ whois example.com
   Domain Name: EXAMPLE.COM
   Registry Domain ID: 2336799_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.iana.org
```

```
┌──(student㉿Kali)-[~]
└─$ whois example.com
   Domain Name: EXAMPLE.COM
   Registry Domain ID: 2336799_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.iana.org
   Registrar URL: http://res-dom.iana.org
   Updated Date: 2026-08-14T08:01:43Z
   Creation Date: 1995-08-14T04:00:00Z
   Registry Expiry Date: 2027-08-13T04:00:00Z
   Registrar: RESERVED-Internet Assigned Numbers Authority
   Registrar IANA ID: 376
   Registrar Abuse Contact Email:
   Registrar Abuse Contact Phone:
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: ELLIOTT.NS.CLOUDFLARE.COM
   Name Server: HERA.NS.CLOUDFLARE.COM
   DNSSEC: signedDelegation
   DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8DD8A46FE06CA230EE23F35B578D64E78B29C3E1C83D245A
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-10-27T12:12:13Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
```

```
┌──(student㉿Kali)-[~]
└─$ nslookup example.com
Server:         10.141.46.119
Address:        10.141.46.119#53

Non-authoritative answer:
Name:   example.com
Address: 104.20.23.154
Name:   example.com
Address: 172.66.147.243
Name:   example.com
Address: 2606:4700:8d75:72db:f2a5:49b:ef6b:ff98


┌──(student㉿Kali)-[~]
└─$ mkdir -p ~/ApexPlanet_Task2/reports

┌──(student㉿Kali)-[~]
└─$ whois example.com > ~/ApexPlanet_Task2/reports/whois_example.txt

┌──(student㉿Kali)-[~]
└─$ nslookup example.com > ~/ApexPlanet_Task2/reports/nslookup_example.txt

┌──(student㉿Kali)-[~]
└─$ ls -lh ~/ApexPlanet_Task2/reports/
total 24K
-rw-rw-r-- 1 student student  895 Oct 22 18:49 ApexPlanet_Task2_nmap_basic.txt
-rw-rw-r-- 1 student student 2.2K Oct 22 18:51 ApexPlanet_Task2_nmap_full.txt
-rw-r--r-- 1 student student  482 Oct 22 18:51 ApexPlanet_Task2_nmap_udp.txt
-rw-rw-r-- 1 student student 1.7K Oct 22 18:50 ApexPlanet_Task2_nmap_versions.txt
-rw-rw-r-- 1 student student  226 Oct 27 17:44 nslookup_example.txt
-rw-rw-r-- 1 student student 3.5K Oct 27 17:43 whois_example.txt

┌──(student㉿Kali)-[~]
└─$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             10.0.2.15/24 fd17:625c:f037:2:1435:f41:38f:24a/64 fd17:625c:f037:2:a00:27ff:fe65:c8
83/64 fe80::a00:27ff:fe65:c883/64
eth1             UP

┌──(student㉿Kali)-[~]
└─$ sudo nmap -sn 192.168.56.0/24 -oN ~/ApexPlanet_Task2/reports/ping_sweep.txt
[sudo] password for student:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-27 17:48 +0530
Nmap scan report for 192.168.56.0
Host is up (0.0050s latency).
Nmap scan report for 192.168.56.1
Host is up (0.0019s latency).
Nmap scan report for 192.168.56.2
```

```
┌──(student㉿Kali)-[~]
└─$ nmap -sV --open 192.168.56.101 -oN ~/ApexPlanet_Task2/reports/banner_grabbing.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-27 17:50 +0530
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.79 seconds

┌──(student㉿Kali)-[~]
└─$ cat ~/ApexPlanet_Task2/reports/banner_grabbing.txt
# Nmap 7.99 scan initiated Tue Oct 27 17:50:50 2026 as: /usr/lib/nmap/nmap --privileged -sV --open -oN /home/studen
t/ApexPlanet_Task2/reports/banner_grabbing.txt 192.168.56.101
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Tue Oct 27 17:50:55 2026 -- 1 IP address (1 host up) scanned in 4.79 seconds

┌──(student㉿Kali)-[~]
└─$ nslookup example.com
Server:         10.141.46.119
Address:        10.141.46.119#53

Non-authoritative answer:
Name:   example.com
Address: 104.20.23.154
Name:   example.com
Address: 172.66.147.243
Name:   example.com
Address: 2606:4700:8d75:72db:f2d7:49c:ef6b:ff98


┌──(student㉿Kali)-[~]
└─$ http://192.168.56.101
zsh: no such file or directory: http://192.168.56.101

┌──(student㉿Kali)-[~]
└─$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 29985 packets, 34M bytes)
num   pkts bytes target     prot opt in     out     source               destination
1        0     0 DROP       tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:80

┌──(student㉿Kali)-[~]
└─$ sudo iptables -L INPUT -n -v > ~/ApexPlanet_Task2/reports/firewall_rules.txt

┌──(student㉿Kali)-[~]
└─$ cat ~/ApexPlanet_Task2/reports/firewall_rules.txt
Chain INPUT (policy ACCEPT 30053 packets, 34M bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DROP       tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:80
```

```
┌──(student㉿Kali)-[~]
└─$ cat ~/ApexPlanet_Task2/reports/firewall_rules.txt
Chain INPUT (policy ACCEPT 30053 packets, 34M bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DROP       tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:80

┌──(student㉿Kali)-[~]
└─$ sudo systemctl status gvmd --no-pager
● gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)
     Loaded: loaded (/usr/lib/systemd/system/gvmd.service; disabled; preset: disabled)
     Active: activating (start) since Tue 2026-10-27 18:04:20 IST; 1s ago
        Job: 17125
 Invocation: 946328b6a59f4c029c3496c61f7e3513
       Docs: man:gvmd(8)
    Process: 73152 ExecStart=/usr/sbin/gvmd --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=_gvm (code=e
xited, status=0/SUCCESS)
      Tasks: 0 (limit: 4618)
     Memory: 32K (peak: 3.7M)
        CPU: 27ms
     CGroup: /system.slice/gvmd.service

Oct 27 18:04:20 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 47.
Oct 27 18:04:20 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 27 18:04:20 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start…irectory
Hint: Some lines were ellipsized, use -l to show in full.

┌──(student㉿Kali)-[~]
└─$ sudo -u postgres psql -l
                                                 List of databases
   Name    |  Owner   | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+---------+-------+--------+-----------+-----------------------
 postgres  | postgres | UTF8     | libc            | en_IN   | en_IN |        |           |
 template0 | postgres | UTF8     | libc            | en_IN   | en_IN |        |           | =c/postgres          +
           |          |          |                 |         |       |        |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | en_IN   | en_IN |        |           | =c/postgres          +
           |          |          |                 |         |       |        |           | postgres=CTc/postgres
(3 rows)


┌──(student㉿Kali)-[~]
└─$ sudo systemctl status gvmd --no-pager
● gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)
     Loaded: loaded (/usr/lib/systemd/system/gvmd.service; disabled; preset: disabled)
     Active: activating (start) since Tue 2026-10-27 18:05:50 IST; 20s ago
        Job: 17344
 Invocation: d9b8dc466731405abb298609d45a0ee7
       Docs: man:gvmd(8)
    Process: 73968 ExecStart=/usr/sbin/gvmd --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=_gvm (code=e
xited, status=0/SUCCESS)
      Tasks: 0 (limit: 4618)
```

```
┌──(student㉿Kali)-[~]
└─$ sudo -u postgres psql -l
                                                List of databases
   Name    |  Owner   | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+---------+-------+--------+-----------+-----------------------
 postgres  | postgres | UTF8     | libc            | en_IN   | en_IN |        |           |
 template0 | postgres | UTF8     | libc            | en_IN   | en_IN |        |           | =c/postgres          +
           |          |          |                 |         |       |        |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | en_IN   | en_IN |        |           | =c/postgres          +
           |          |          |                 |         |       |        |           | postgres=CTc/postgres
(3 rows)


┌──(student㉿Kali)-[~]
└─$ sudo journalctl -u gvmd -n 60 --no-pager
Oct 27 17:53:46 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 27 17:53:46 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 27 17:53:46 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 27 17:53:46 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 40.
Oct 27 17:53:46 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 27 17:53:46 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such
 file or directory
Oct 27 17:55:17 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 27 17:55:17 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 27 17:55:17 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 27 17:55:17 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 41.
Oct 27 17:55:17 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 27 17:55:17 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such
 file or directory
Oct 27 17:56:47 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 27 17:56:47 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 27 17:56:47 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 27 17:56:47 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 42.
Oct 27 17:56:47 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 27 17:56:47 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such
 file or directory
Oct 27 17:58:18 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 27 17:58:18 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 27 17:58:18 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 27 17:58:18 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 43.
Oct 27 17:58:18 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
Oct 27 17:58:18 Kali systemd[1]: gvmd.service: Can't open PID file '/run/gvmd/gvmd.pid' (yet?) after start: No such
 file or directory
Oct 27 17:59:48 Kali systemd[1]: gvmd.service: start operation timed out. Terminating.
Oct 27 17:59:48 Kali systemd[1]: gvmd.service: Failed with result 'timeout'.
Oct 27 17:59:48 Kali systemd[1]: Failed to start gvmd.service - Greenbone Vulnerability Manager daemon (gvmd).
Oct 27 17:59:48 Kali systemd[1]: gvmd.service: Scheduled restart job, restart counter is at 44.
Oct 27 17:59:48 Kali systemd[1]: Starting gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)...
```

```
┌──(student㉿Kali)-[~]
└─$ sudo systemctl stop gvmd

┌──(student㉿Kali)-[~]
└─$ sudo -u postgres psql -c "\dx" postgres
                        List of installed extensions
  Name   | Version | Default version |   Schema   |         Description
---------+---------+-----------------+------------+------------------------------
 plpgsql | 1.0     | 1.0             | pg_catalog | PL/pgSQL procedural language
(1 row)


┌──(student㉿Kali)-[~]
└─$ sudo grep -nE 'db_address|db_port|db_name|db_user' /etc/gvm/gvmd.conf

┌──(student㉿Kali)-[~]
└─$ apt-cache show gvmd | grep -E '^(Package|Version|Depends|Description):'
Package: gvmd
Version: 26.24.0-1
Depends: adduser, doc-base, greenbone-feed-sync (≥ 24.1.1), gvmd-common (= 26.24.0-1), notus-scanner (≥ 22.4.0),
ospd-openvas (≥ 22.9.1-2), xml-twig-tools, postgresql-18, postgresql-18-pg-gvm, libbsd0 (≥ 0.6.0), libc6 (≥ 2.38
), libcjson1 (≥ 1.7.13), libglib2.0-0t64 (≥ 2.76.0), libgnutls30t64 (≥ 3.7.5), libgpgme45 (≥ 1.23.2), libgvm22t
64 (≥ 22.39.0), libical3t64 (≥ 3.0.0), libpq5
Description: Manager Module of Greenbone Vulnerability Manager

┌──(student㉿Kali)-[~]
└─$ ls /usr/share/doc/ | grep -Ei 'gvm|greenbone|openvas'
greenbone-feed-sync
gvm
gvmd
gvmd-common
libgvm22t64
openvas-scanner
ospd-openvas
postgresql-18-pg-gvm
python3-gvm

┌──(student㉿Kali)-[~]
└─$ command -v greenbone-feed-sync
/usr/bin/greenbone-feed-sync

┌──(student㉿Kali)-[~]
└─$ greenbone-feed-sync --help
usage: greenbone-feed-sync [--print-completion {bash,zsh,tcsh}] [--version] [-h] [--selftest] [--verbose |
                           --quiet] [-c CONFIG] [--private-directory PRIVATE_DIRECTORY]
                           [--compression-level {0,1,2,3,4,5,6,7,8,9}]
                           [--type {all,nvt,gvmd-data,scap,cert,notus,nasl,report-format,scan-config,port-list}]
                           [--feed-version FEED_VERSION] [--feed-release FEED_RELEASE]
                           [--destination-prefix DESTINATION_PREFIX]
```

```
┌──(student㉿Kali)-[~]
└─$ gvmd --help | grep -i -E 'create|migrate|database|role'
  --create-encryption-key                    Create a new credential encryption key, set it as the new default
 and exit. With no other options given, a 4096 bit RSA key is created.
  --create-scanner=<scanner>                 Create global scanner <scanner> and exit.
  --create-user=<username>                   Create admin user <username> and exit.
  -d, --database=<name>                      Use <name> as database for PostgreSQL.
  --db-host=<host>                           Use <host> as database host or socket directory for PostgreSQL.
  --db-port=<port>                           Use <port> as database port or socket extension for PostgreSQL.
  --db-user=<user>                           Use <user> as database user.
  --get-roles                                List roles and exit.
  --max-database-connections=<number>        Maximum number of database connections at the same time. Default:
 50
  -m, --migrate                              Migrate the database and exit.
  --password=<password>                      Password, for --create-user.
  --role=<role>                              Role for --create-user and --get-users.
  --scanner-ca-pub=<scanner-ca-pub>          Scanner CA Certificate path for --[create|modify]-scanner.
  --scanner-credential=<scanner-credential>  Scanner credential for --create-scanner and --modify-scanner. Can
 be blank to unset or a credential UUID. If omitted, a new credential can be created instead.
  --scanner-host=<scanner-host>              Scanner host or socket for --create-scanner and --modify-scanner.
  --scanner-key-priv=<scanner-key-private>   Scanner private key path for --[create|modify]-scanner if --scann
er-credential is not given.
  --scanner-key-pub=<scanner-key-public>     Scanner Certificate path for --[create|modify]-scanner if --scann
er-credential is not given.
  --scanner-port=<scanner-port>              Scanner port for --create-scanner and --modify-scanner. Default i
s 9390.
  --scanner-relay-host=<scanner-relay-host>  Scanner relay host or socket for --create-scanner and --modify-sc
anner.
  --scanner-relay-port=<scanner-relay-port>  Scanner relay port for --create-scanner and --modify-scanner.
  --scanner-type=<scanner-type>              Scanner type for --create-scanner and --modify-scanner. Either 'O
penVAS', 'OSP', 'OSP-Sensor' or a number as used in GMP.
  --no-default-certs                         Bypass reading/validating scanner default certificate files for -
-create-scanner.
  --secinfo-commit-size=<number>             During CERT and SCAP sync, commit updates to the database every <
number> items, 0 for unlimited, default: 0

┌──(student㉿Kali)-[~]
└─$ ls -l /usr/share/postgresql/18/extension/pg-gvm*
-rw-r--r-- 1 root root 1181 Mar  9  2026 /usr/share/postgresql/18/extension/pg-gvm--1.0--1.1.sql
-rw-r--r-- 1 root root  139 Mar  9  2026 /usr/share/postgresql/18/extension/pg-gvm--1.1--22.4.0.sql
-rw-r--r-- 1 root root  139 Mar  9  2026 /usr/share/postgresql/18/extension/pg-gvm--22.4.0--22.5.sql
-rw-r--r-- 1 root root  139 Mar  9  2026 /usr/share/postgresql/18/extension/pg-gvm--22.5--22.6.sql
-rw-r--r-- 1 root root 2973 Mar 16  2026 /usr/share/postgresql/18/extension/pg-gvm--22.6.sql
-rw-r--r-- 1 root root   94 Mar 16  2026 /usr/share/postgresql/18/extension/pg-gvm.control

┌──(student㉿Kali)-[~]
└─$ sudo find /usr/share/doc/gvmd /usr/share/doc/postgresql-18-pg-gvm -maxdepth 2 -type f 2>/dev/null
/usr/share/doc/gvmd/changelog.Debian.gz
/usr/share/doc/gvmd/copyright
```

```
┌──(student㉿Kali)-[~]
└─$ zcat /usr/share/doc/gvmd/README.md.gz | head -80
```

```
![Greenbone Logo](https://www.greenbone.net/wp-content/uploads/gb_new-logo_horizontal_rgb_small.png)

# Greenbone Vulnerability Manager <!-- omit in toc -->

[![GitHub releases](https://img.shields.io/github/release/greenbone/gvmd.svg)](https://github.com/greenbone/gvmd/releases)
[![Build and Test](https://github.com/greenbone/gvmd/actions/workflows/build-and-test.yml/badge.svg)](https://github.com/greenbone/gvmd/actions/workflows/build-and-test.yml)

The Greenbone Vulnerability Manager is the central management service between
security scanners and the user clients.

It manages the storage of any vulnerability management configurations and of the
scan results. Access to data, control commands and workflows is offered via the
XML-based Greenbone Management Protocol (GMP). Controlling scanners like
the [OPENVAS Scanner](https://github.com/greenbone/openvas-scanner)* is done via
the Open Scanner Protocol (OSP).

- [Releases](#releases)
- [Installation and Usage](#installation-and-usage)
- [Support](#support)
- [Maintainer](#maintainer)
- [Contributing](#contributing)
  - [Code style and formatting](#code-style-and-formatting)
  - [CMake formatting](#cmake-formatting)
- [License](#license)

## Releases

All [release files](https://github.com/greenbone/gvmd/releases) are signed with
the [Greenbone Community Feed integrity key](https://community.greenbone.net/t/gcf-managing-the-digital-signatures/101).
This gpg key can be downloaded at https://www.greenbone.net/GBCommunitySigningKey.asc
and the fingerprint is `8AE4 BE42 9B60 A59B 311C  2E73 9823 FAA6 0ED1 E580`.

## Installation and Usage

This module can be configured, built and installed with following commands:

```sh
cmake .
make install
```

For detailed installation requirements and instructions, please see the file
```

```
┌──(student㉿Kali)-[~]
└─$ zcat /usr/share/doc/gvmd/changelog.Debian.gz | head -60
gvmd (26.24.0-1) unstable; urgency=medium

  * New upstream version 26.24.0
  * Refresh patches

 -- Sophie Brun <sophie@kali.org>  Tue, 28 Apr 2026 15:50:04 +0200

gvmd (26.23.0-1) unstable; urgency=medium

  * New upstream version 26.23.0
  * Refresh patches

 -- Sophie Brun <sophie@kali.org>  Mon, 20 Apr 2026 15:13:21 +0200

gvmd (26.19.0-1) unstable; urgency=medium

  * New upstream version 26.19.0
  * Refresh patches

 -- Sophie Brun <sophie@kali.org>  Mon, 16 Mar 2026 14:57:34 +0100

gvmd (26.16.1-1) unstable; urgency=medium

  * New upstream version 26.16.1
  * Update debian/copyright
  * Refresh patches
  * Bump Standards-Version to 4.7.3

 -- Sophie Brun <sophie@kali.org>  Tue, 24 Feb 2026 17:27:22 +0100

gvmd (26.15.0-1) unstable; urgency=medium

  * New upstream version 26.15.0
  * Refresh patches

 -- Sophie Brun <sophie@kali.org>  Fri, 30 Jan 2026 16:30:53 +0100

gvmd (26.14.0-1) unstable; urgency=medium

  * New upstream version 26.14.0
  * Refresh patches

 -- Sophie Brun <sophie@kali.org>  Thu, 22 Jan 2026 17:04:17 +0100

gvmd (26.13.0-1) unstable; urgency=medium
```

```
┌──(student㉿Kali)-[~]
└─$ sudo  -u postgres psql -c "SELECT name, default_version, installed_version FROM pg_available_extensions WHERE n
ame = 'pg-gvm'; postgres
dquote>

┌──(student㉿Kali)-[~]
└─$ sudo  -u postgres psql -c "SELECT name, default_version, installed_version FROM pg_available_extensions WHERE n
ame = 'pg-gvm'; postgres

dquote>

┌──(student㉿Kali)-[~]
└─$ sudo -u postgres psql -l
[sudo] password for student:
                                                List of databases
   Name    |  Owner   | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+---------+-------+--------+-----------+-----------------------
 postgres  | postgres | UTF8     | libc            | en_IN   | en_IN |        |           |
 template0 | postgres | UTF8     | libc            | en_IN   | en_IN |        |           | =c/postgres          +
           |          |          |                 |         |       |        |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | en_IN   | en_IN |        |           | =c/postgres          +
           |          |          |                 |         |       |        |           | postgres=CTc/postgres
(3 rows)


┌──(student㉿Kali)-[~]
└─$ command -v gvm-setup

┌──(student㉿Kali)-[~]
└─$ command -v gvm-check-setup

┌──(student㉿Kali)-[~]
└─$ dpkg -L gvmd | grep -Ei 'setup|REAME|migration
pipe quote> student
pipe quote> student
pipe quote> student
pipe quote>

┌──(student㉿Kali)-[~]
└─$ dpkg -L gvmd
/.
/etc
/etc/gvm
/etc/gvm/gvmd.conf
/etc/gvm/gvmd_log.conf
/etc/gvm/pwpolicy.conf
/etc/logrotate.d
/etc/logrotate.d/gvmd
/usr
```


┌──(student㉿Kali)-[~]
└─$ cat /etc/gvm.gvmd.cinf
cat: /etc/gvm.gvmd.cinf: No such file or directory
┌──(student㉿Kali)-[~]
└─$ cat /etc/gvm/gvmd.conf
[authentication]
jwt_access_duration = 60
jwt_secret_type = ecdsa
jwt_decode_secret_path = /var/lib/gvm/gvmd/jwt/ecdsa-public.pem
jwt_encode_secret_path = /var/lib/gvm/gvmd/jwt/ecdsa-private.pem
jwt_decode_secret = <INSERT_SECRET_HERE>
jwt_encode_secret = <INSERT_SECRET_HERE>
[features]
enable_agents = false
enable_container_scanning = false
enable_credential_store = false
enable_openvasd = false
enable_vt_metadata = false
enable_security_intelligence_export = false
enable_jwt_auth = false
[security_intelligence_export]
max_retries = 10
┌──(student㉿Kali)-[~]
└─$ dpkg -L postgresql-18-pg-gvm
/.
/usr
/usr/lib
/usr/lib/postgresql
/usr/lib/postgresql/18
/usr/lib/postgresql/18/lib
/usr/lib/postgresql/18/lib/libpg-gvm.so
/usr/share
/usr/share/doc
/usr/share/doc/postgresql-18-pg-gvm
/usr/share/doc/postgresql-18-pg-gvm/changelog.Debian.gz
/usr/share/doc/postgresql-18-pg-gvm/copyright
/usr/share/postgresql
/usr/share/postgresql/18
/usr/share/postgresql/18/extension
/usr/share/postgresql/18/extension/pg-gvm--1.0--1.1.sql
/usr/share/postgresql/18/extension/pg-gvm--1.1--22.4.0.sql
/usr/share/postgresql/18/extension/pg-gvm--22.4.0--22.5.sql
/usr/share/postgresql/18/extension/pg-gvm--22.5--22.6.sql
/usr/share/postgresql/18/extension/pg-gvm--22.6.sql

```
┌──(student㉿Kali)-[~]
└─$ cat /etc/gvm.gvmd.cinf
cat: /etc/gvm.gvmd.cinf: No such file or directory

┌──(student㉿Kali)-[~]
└─$ cat /etc/gvm/gvmd.conf
[authentication]
jwt_access_duration = 60
jwt_secret_type = ecdsa
jwt_decode_secret_path = /var/lib/gvm/gvmd/jwt/ecdsa-public.pem
jwt_encode_secret_path = /var/lib/gvm/gvmd/jwt/ecdsa-private.pem
# jwt_decode_secret = <INSERT_SECRET_HERE>
# jwt_encode_secret = <INSERT_SECRET_HERE>

[features]
enable_agents = false
enable_container_scanning = false
enable_credential_store = false
enable_openvasd = false
enable_vt_metadata = false
enable_security_intelligence_export = false
enable_jwt_auth = false

[security_intelligence_export]
max_retries = 10

┌──(student㉿Kali)-[~]
└─$ dpkg -L postgresql-18-pg-gvm
/.
/usr
/usr/lib
/usr/lib/postgresql
/usr/lib/postgresql/18
/usr/lib/postgresql/18/lib
/usr/lib/postgresql/18/lib/libpg-gvm.so
/usr/share
/usr/share/doc
/usr/share/doc/postgresql-18-pg-gvm
/usr/share/doc/postgresql-18-pg-gvm/changelog.Debian.gz
/usr/share/doc/postgresql-18-pg-gvm/copyright
/usr/share/postgresql
/usr/share/postgresql/18
/usr/share/postgresql/18/extension
/usr/share/postgresql/18/extension/pg-gvm--1.0--1.1.sql
/usr/share/postgresql/18/extension/pg-gvm--1.1--22.4.0.sql
/usr/share/postgresql/18/extension/pg-gvm--22.4.0--22.5.sql
/usr/share/postgresql/18/extension/pg-gvm--22.5--22.6.sql
/usr/share/postgresql/18/extension/pg-gvm--22.6.sql
```

10. Conclusion

The Nmap scan successfully discovered a live Metasploitable2 host and 23 open TCP ports, with a range of network services and legacy software banners. These results provide a baseline for analysing network exposure in the lab. OS detection was inconclusive. Wireshark evidence and final firewall rules require confirmation from the saved files, and the Greenbone vulnerability report remains pending because the manager did not start successfully.